

Ethan Joseph

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EDUCATION

Bachelor of Engineering (B. Eng) – Software Engineering  
McMaster University

Graduation Date: June 2030

WORK EXPERIENCE

- Software Developer - Zivko Online SolutionsMay 2026 – August 2026
- Owned and shipped a **full-stack analytics workflow for Zivko Online Solutions**, translating stakeholder requirements into production features across an existing **React, Node.js, and Supabase** codebase.
  - Built a server-side data pipeline integrating **4 independent systems: Google Search Console, SE Ranking, WordPress, and Supabase**, with URL normalization and structured aggregation to unify fragmented third-party data into a single application workflow.
  - Redesigned third-party API orchestration around **on-demand execution**, reducing third-party API calls by **~65%** through **24-hour caching, request deduplication, rate-limit retries, and controlled request sequencing**.
  - Developed a **decision and recommendation layer** combining **5+ independent signal types** across performance, ranking, authority, backlink, and audit data into prioritized application outputs, with fault-tolerant handling that preserved usable results when individual upstream services failed.
  - Partnered directly with **project stakeholders and a 3-person development team** through feature demos, technical feedback, regression testing, Git-based review, and production handoff, ultimately shipping the workflow to the live application.
- Founder & Organizer - Eclipse Hackathons - [View Devpost](#)August 2024 – November 2024
- Founded and led an **80+ participant hackathon across 15+ Ontario schools**, securing HCLTech sponsorship and recruiting judges from AMD, Dropbox, Fractal Analytics, and SAP Concur.
  - Designed and developed the event’s website, registration system, submission portal, automated email workflows, and backend application infrastructure supporting **80+ registered users with zero downtime during live judging**.

PROJECTS

- Sentinel Security Auditor – [View Live Site](#) | Python, TypeScript, Gemini API, GitHub Actions, VercelNov 2025 – Present
- Built an AI-driven application security auditor that analyzes local and GitHub repositories to generate structured vulnerability findings, attack-flow graphs, and remediation guidance.
  - Deployed a production web interface on Vercel, averaging **~800 monthly visits** from users scanning their repositories for security vulnerabilities.
  - Developed a pipeline to analyze large codebases (**50k+ LOC**) by using prompt compression techniques, reducing token spend by 28%.
  - Developed repository-aware vulnerability analysis that **reduced false positives by ~35%** vs rule-only scanners (benchmarked across 40+ open-source repos).
  - Implemented automated remediation with structured Git patch generation, achieving **~70% successful fix** application rate as measured by post-scan verification and rescanning of patched code.
- SafeGuard - [MacHacks Winner: Best Use of Gemini API](#) | JavaScript, React, Tailwind CSS, Flask, Gemini API, VercelMar 2026
- Built and deployed an AI safety application that analyzes environmental images to identify hazards and generate structured risk reports.
  - Engineered a Gemini vision pipeline with normalized JSON outputs supporting **up to 8 images per request** and context-aware follow-up analysis.
  - Implemented client-side image compression and serverless API processing, **reducing average payload size by ~60%** before inference.

TECHNICAL SKILLS

Languages | Python, TypeScript/JavaScript, Java, SQL, C/C++  
Frameworks & Technologies | React, Node.js, Flask, Supabase, REST API Development, Serverless APIs  
AI Engineering | Gemini API, LLM Application Development, Structured Outputs, Prompt & Context Optimization, Multi-API Data Integration  
Systems & Developer Tools | Git, GitHub Actions, Docker, Linux, Vercel, CI/CD, Caching, Rate-Limit Handling, Unit Testing, Debugging